

Relational calculus in algebra and logic

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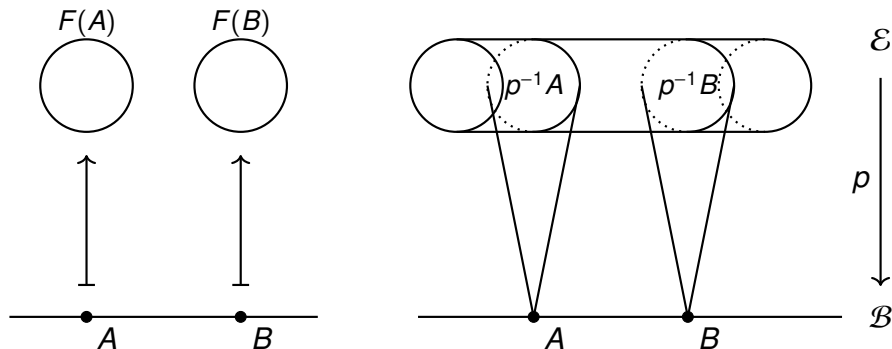
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Grothendieck fibrations in algebra and logic

The **Grothendieck construction** yields an equivalence

$$[\mathcal{C}^{\text{op}}, \text{Pos}] \simeq \text{FaithAmnFib}(\mathcal{C})$$

between **poset-valued presheaves** (a key tool in **categorical logic**) and **faithful amnestic fibrations** (a key tool in **categorical algebra**).



Definition (Z. Janelidze).

A **form** on $\mathcal{C}$ is a faithful amnestic functor $F: \mathcal{B} \rightarrow \mathcal{C}$,
or equivalently **a functor F whose fibers are posets**.

Idea and motivation:

- **self-dual axiomatic framework for algebra**
- **it abstracts the bifibration of subobjects** (or of **quotients**)
over an abelian category

Further axioms:

Orean form: **bifibration whose fibers are bounded lattices**

Noetherian form: further self-dual constraints which yield **Noether's isomorphism theorems and diagram lemmas**

Definition (W. Lawvere).

A **doctrine** on a category $\mathcal{C}$ is a functor $F: \mathcal{C}^{\text{op}} \rightarrow \mathcal{Pos}$.

Idea and motivation:

- **it abstract the relationship between the syntax of a logical theory** (types, contexts, and terms) **and its semantics** (predicates, formulas, and proofs)
- **substitution is modeled by change of base**
- quantifiers as adjoints to substitution along projections

Further axioms:

primary, elementary, existential: **additional structure on and between the fibers to model connectives**

Relational doctrines

Definition (F. Dagnino – F. Pasquali).

A **relational doctrine** on $\mathcal{C}$ is a functor $R : (\mathcal{C} \times \mathcal{C})^{\text{op}} \rightarrow \mathcal{Pos}$ with

- an **identity** relation $d_X \in R(X, X)$ for every X in $\mathcal{C}$
- a monotone **composition** function
 $-; - : R(X, Y) \times R(Y, Z) \rightarrow R(X, Z)$,
for every X, Y, Z in $\mathcal{C}$ such that $R_{f,g}(\alpha) ; R_{g,h}(\beta) \leq R_{f,h}(\alpha ; \beta)$
- a monotone **converse** function $(-)^{\perp} : R(X, Y) \rightarrow R(Y, X)$, for
every X, Y in $\mathcal{C}$ such that $(R_{f,g}(\alpha))^{\perp} \leq R_{g,f}(\alpha^{\perp})$

satisfying appropriate equations.

Idea and motivation:

it abstracts the assignment of the poset of relations from X
to Y on every $(X, Y) \in \mathcal{C} \times \mathcal{C}$

Definition (Z. Janelidze – D. Rodelo).

An **arc** (abstract relational calculus) is a **poset-enriched category** equipped with an **involution**.

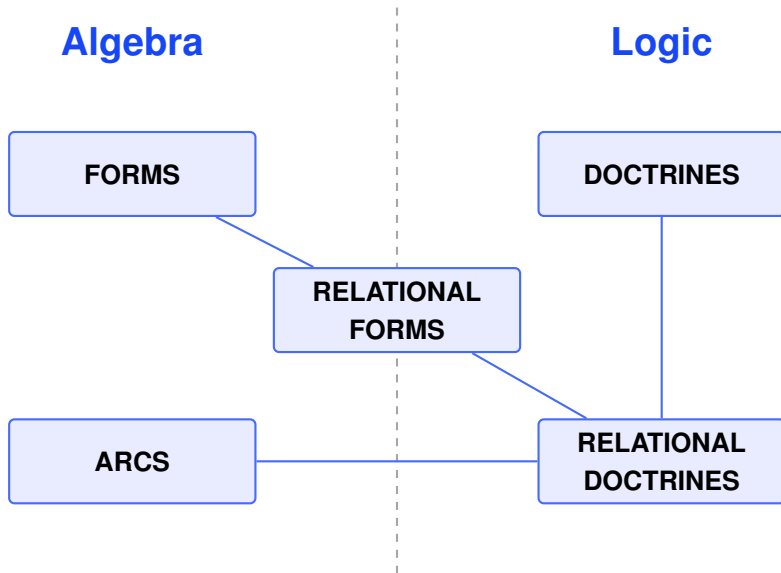
We can represent an arc as a triple $(\mathcal{C}, \leq, \circ)$, where

- $\leq$ is a partial order on parallel morphisms in $\mathcal{C}$ such that composition of morphisms is monotone in each argument
- $\circ$ is an isomorphism $\mathcal{C} \rightarrow \mathcal{C}^{\text{op}}$ which is identity on objects and preserves the partial order

Idea and motivation:

- **study relational calculi of Barr exact Mal'tsev categories**
- **universe of relations** (syntax and semantics are in the same place rather than in two different levels).

The key players



Properties via relations

We can **formulate properties and structures in terms of the (generalized) relations** given by a relational doctrine (or an arc).

α is *injective*: $\alpha; \alpha^\perp \leq d_X$

α is *surjective*: $d_Y \leq \alpha^\perp; \alpha$

α is *reflexive*: $X = Y$ and $d_X \leq \alpha$

α is *functional*: $\alpha^\perp; \alpha \leq d_Y$

α is *total*: $d_X \leq \alpha; \alpha^\perp$

α is *symmetric*: $X = Y$ and $\alpha^\perp \leq \alpha$

α is *transitive*: $X = Y$ and $\alpha; \alpha \leq \alpha$

α is a *congruence*: $\alpha = \beta^\perp \beta$ for a functional relation β

Arcs are particular relational doctrines

Definition.

A relational doctrine R is **extensional** if for every f, g in $\mathcal{C}$, $d_X \leq R_{f,g}(d_Y)$ implies $f = g$.

Definition.

R satisfies the **rule of unique choice (RUC)** if, for every functional and total relation $\alpha \in R(X, Y)$, there is an arrow $f : X \rightarrow Y$ in $\mathcal{C}$ s.t. $\Gamma_f \leq \alpha$.

Theorem (F. Dagnino – F. Pasquali).

Arcs are precisely relational doctrines which satisfy extensionality and rule of unique choice.

Extending regularity from arcs to relational doctrines

Theorem (Z. Janelidze – D. Rodelo).

For an arc, the following conditions are equivalent:

- (R1) every functional relation factorizes as a surjection followed by an injection*
- (R2) every congruence is the kernel of some surjective relation*
- (R3) every cocongruence is the opkernel of some injective relation*
- (R4) **surjective functional relations and injective functional relations form a (proper) factorisation system***
- (R5) **every congruence C has a surjective quotient***
- (R6) every cocongruence C has an injective insertion*

Definition.

An arc is **regular** if it satisfies any of the previous equivalent conditions.

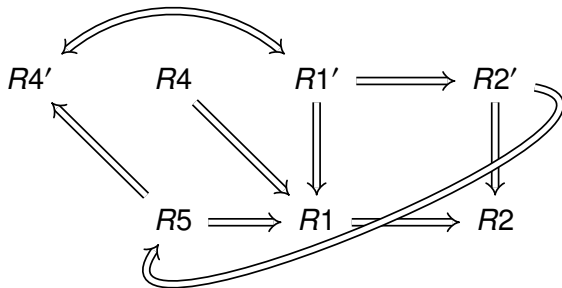
Extending regularity from arcs to relational doctrines

Definition.

A relational doctrine is **regular** if it satisfies **both**:

(R5) **every congruence has a surjective effective quotient**

(R6) every cocongruence has an injective full comprehension



where **primes upgrade surjections to surjective effective quotients** and injections to injective full comprehensions.

Extending exactness from arcs to relational doctrines

Definition.

A relational doctrine is **exact** if it satisfies **both**:

- (E1') **every equivalence relation is the kernel of a surjective effective quotient**
- (E2') every coequivalence relation is the image of an injective full comprehension

Proposition.

*A relational doctrine is exact if and only if it is **regular** and such that **every equivalence relation is a congruence** and every coequivalence relation is a cocongruence.*

Extending Mal'tsev from arcs to relational doctrines

Theorem (Z. Janelidze – D. Rodelo).

*For an exact arc, the following are equivalent and define **Mal'tsev**:*

- (M1) *Every accessible reflexive relation is a congruence and every accessible coreflexive relation is a cocongruence;*
- (M2) *every accessible reflexive relation is symmetric and every accessible coreflexive relation is symmetric;*
- (M3) *given $S_1, S_2 \in R(X, X)$ accessible and symmetric, if they are both reflexive or both coreflexive then $S_1; S_2 = S_2; S_1$;*
- (M4) *composite of congruences is a congruence and composite of cocongruences is a cocongruence;*
- (M5) *every accessible relation is the composite of a surjective functional relation and an injective total relation;*
- (M6) *every accessible relation is difunctional.*

Extending Mal'tsev from arcs to relational doctrines

Theorem.

Let R be an exact relational doctrine. Then

$$M5 \Rightarrow M6 \Rightarrow M1 \Rightarrow M2 \Rightarrow M3 \Rightarrow M4$$

*Moreover, **if R satisfies stronger conditions, like noetherianity, then $M4 \Rightarrow M5$, and hence the conditions are all equivalent.***

Transporting results from logic to algebra

We can transport the quotient and comprehension completions of a relational doctrine to the framework of arcs.

Note: **the quotient completion does not distribute over the comprehension completion nor vice versa** in general, but we can identify sufficient conditions.

To partially solve the non-distributivity, we produced a **splitting of symmetric idempotents completion for arcs**.

Relational forms: the idea

Forms	Relational Forms
Subobject (or quotient) Bifibration of subobjects	Endorelation Bifibration of congruences

Idea: **Relational forms host algebra while being close to logic and thus being able to draw from logical techniques.**

Perks: **The structure is significantly simpler than that of relational doctrines.** This is because endorelations on a given set X form an **ordered involutive monoid**:

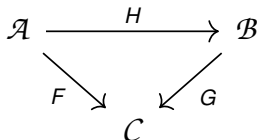
- the product is composition of endorelations
- the unit is the diagonal relation
- the involution is the converse of relations

The category of forms

We denote as **$\mathcal{F}orms(\mathcal{C})$** the category of forms over $\mathcal{C}$:

an object is a form over $\mathcal{C}$,

a morphism $F \rightarrow G$ is a commutative triangle in $\mathcal{C}at$



$\mathcal{F}orms(\mathcal{C})$ has a cartesian monoidal structure: the tensor is given by **pullbacks** and the unit is the identity functor.

Relational forms

Definition.

A **relational form** on $\mathcal{C}$ is an **involutive monoid** in the category $\mathcal{F}orms(\mathcal{C})$ which is also a bifibration.

Note: For every f we have an adjunction $f_! \dashv f^*$. This induces both a **monad** $f^* \circ f_!$ and a **comonad** $f_! \circ f^*$.

The idea is to **use these monads and comonads to substitute the meets and joins** of the theory of forms.

Theorem.

Let $R: (\mathcal{C} \times \mathcal{C})^{\text{op}} \rightarrow \mathcal{P}os$ be a relational doctrine. Then the Grothendieck construction of the precomposition $R \circ \Delta^{\text{op}}$ of R with the op of the diagonal $\Delta: \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$ is a relational form.

Kernels and images, quotients and comprehensions

Definition.

Let $F: \mathcal{A} \rightarrow \mathcal{C}$ be a relational form over $\mathcal{C}$, and let $f: X \rightarrow Y$ in $\mathcal{C}$.

$$\begin{array}{ccc} \text{ker}(f) = f^*(d_Y) & \xrightarrow{\bar{f}^{d_Y}} & d_Y \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array} \qquad \begin{array}{ccc} d_X & \xrightarrow{d_X \bar{f}} & f_!(d_X) = \text{im}(f) \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

Restricting F to the kernels yields a form F_n over $\mathcal{C}$ that is a fibration. The unit $d: \mathcal{C} \rightarrow \mathcal{A}$ becomes a fibred initial object for F_n .

Theorem.

Let $F: \mathcal{A} \rightarrow \mathcal{C}$ be a relational form. The following are equivalent:

- (i) F has **quotients of kernels**
- (ii) the restriction $d: \mathcal{C} \rightarrow \mathcal{A}_n$ of d to the kernels **has a left adjoint**.

Noetherian relational forms

Definition.

A relational form $F : \mathcal{A} \rightarrow \mathcal{C}$ is **Noetherian** if it satisfies the axioms:

regularity F has all quotients of kernels and all comprehensions of images

balancing bijective arrows in $\mathcal{C}$ are exactly the isomorphisms

stability for every f in $\mathcal{C}$, $f^* \circ f_!$ preserves kernels and $f_! \circ f^*$ preserves images

composition $f^* \circ f_!(K) = \ker(f); K; \ker(f)$, $f_! \circ f^*(C) = \text{im}(f); C; \text{im}(f)$
for every f in $\mathcal{C}$, K kernel and C image

Conjecture.

Noetherian relational forms satisfy all homological diagram lemmas and Noether's isomorphism theorems.

Some results on Noetherian relational forms

Let $F: \mathcal{A} \rightarrow \mathcal{C}$ be a **Noetherian relational form** over $\mathcal{C}$.

Proposition.

- For every $X \in \mathcal{C}$, *the poset of kernels over X has finite joins, and the poset of images over X has finite meets*
- All *pushouts of quotients* and pullbacks of comprehensions exist
- *The pyramid construction can be performed*

Theorem (The first homomorphism theorem).

Every morphism f in $\mathcal{C}$ factorizes as a quotient followed by a comprehension.

$$\begin{array}{ccccc} \ker(f) & & & & \text{im}(f) \\ \Downarrow & & & & \Downarrow \\ X & \xrightarrow{\quad f \quad} & & & Y \\ & \searrow q_f & Q \cong I & \nearrow i_f & \\ & & W & & \end{array}$$